Demultiplexer (Demux)

- takes a single input & distributes it over several outputs.

ex: 1 line-to-8-line demultiplexer

1 input, 3 selectors, 8 outputs

* random disclaimer!:

to add two n-bit numbers together, n full-adders should be cascaded.

Label gate outputs

1 - label all gate outputs that are a function of input variables

2 - label gates that are a function of input variables & previously labeled gates

3 - repeat process until all outputs are labeled

Binary multiplier

B_1 B_0

A_1 A_0

A_0B_1 A_0B_0 → output

A_1B_1 A_1B_0

P_3 P_2 P_1 P_0 → into adder

carry out, sum (output)

into adder

carry out, sum

Chapter 5 - Synchronous sequential logic

- digital system ⌊ registers ⌊ data processing units
- Binary info is transferred from one set of registers into another
- transfer can be direct or via a processing unit to perform an operation.

register

- group of Binary cells
- n cells can store n bits
- can be in 2^n states

Sequential logic circuits

1 - The input variables

2 - the previous states (internal states).

1 - time 2 - memory element

Flip-flop

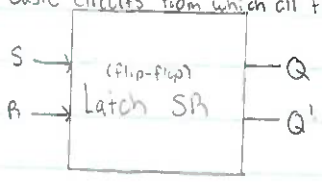
- memory cell that stores one "bit" of info
- a set of these can be used to store several bits in a register.

2 different modes ⌊ asynchronous ⌊ output reacts "immediately" to the changes of the input variables

⌊ synchronous ⌊ changes of the input variables will become effective when the clock input is active

Latches

- Basic circuits from which all flip-flops are constructed



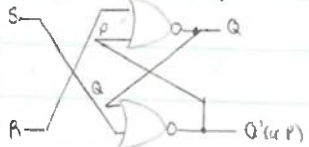
S is for set

R is for reset

Q' (complement state)

Q (normal state)

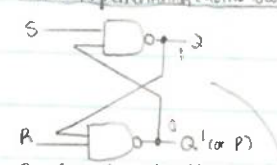
NOR implementation (active-high)



$Q = (R+P)' = R'P$

$P = (S+Q)' = S'Q'$

NAND implementation (active low)



$Q = (Q'S)' = (PS)'$

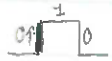
$Q' = (RQ)'$

clock signal

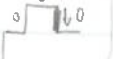
- positive level trigger



- positive transition trigger (positive edge)



- negative transition trigger (negative edge)



S	R	Q	Q'	
1	0	1	0	
0	1	0	1	
0	0	1	0	after SR: 10
0	0	0	1	after SR: 01
1	1	?	?	not allowed
1	1	?	?	10 01

S	R	Q	Q'	
1	0	0	1	
0	1	1	0	
0	0			(not allowed)
0	0			
1	1	0	1	after SR: 10
1	1	1	0	after SR: 01